

16. BRICK AND BLOCK LAYING

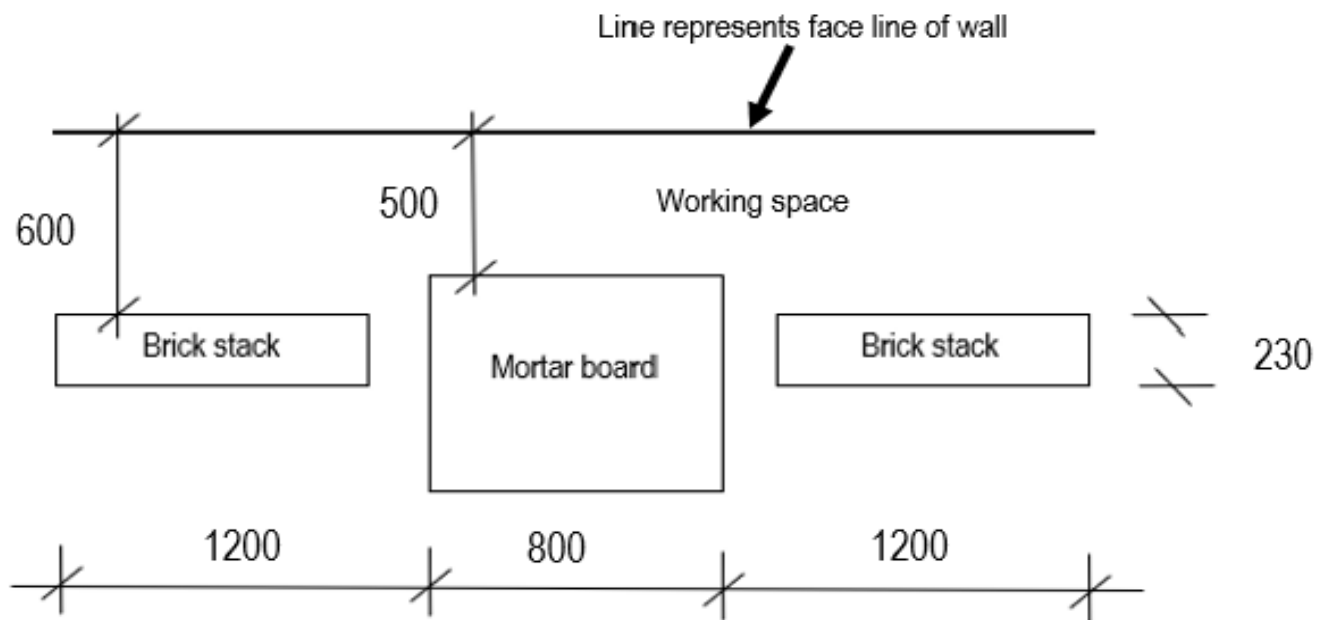
Bricks and blocks are made to last. The average lifespan of a building made from bricks is estimated at more than 100 years. Their lifespan and limited environmental impact make bricks unbeatable if you want to build a durable building. Bricks make it possible to create beautiful buildings with limited costs and a long lifespan. They do not rot, dent, or need to be painted. And unlike other materials, bricks often look better with age.

In general, New Place Construction will engage a qualified Bricklaying sub-contractor to complete bricklaying tasks, but you may be asked to assist so the following will guide you in general brick and block laying procedures.

Preparing to Lay Bricks and Blocks

Like all construction activities, planning the task prior to commencing is an important process that must be conducted, as it increases efficiency and safety and decreases costs. In general, for brick and block laying activities, a New Place Construction employee may be required to:

- Follow any planning documents (Work Plan, JSEA, Drawings, Australian Standards Codes of Conduct)
- Follow operator manuals to perform pre-operational checks on power cords, tools and equipment, which may include:
 - reading the manufacturer's instructions (including parts, uses, limitations, cleaning and troubleshooting)
 - checking lubricants, water, guards, controls and switches
 - checking and adjusting clamps, guides used for holding and supporting materials
 - checking leads for damage and tags
 - connect power equipment to an RCD
 - read the SWMS and SOP before switching on the power.
 - wear PPE for hearing and respiratory protection
- Inspect, identify and record worksite hazards, which may include:
 - trip hazards
 - manual handling hazards
 - tool and equipment hazards
 - environmental hazards
- Prepare hazards and risk controls, which may include:
 - signage and barricades
 - preparing bins
 - preparing wash down area
 - preparing PPE
 - preparing tools and equipment
- Safely prepare and position brick and block materials for use, which may include:
 - identifying a flat working location, close to the brick or block wall (eg 600mm) and mortar boards
 - clearing any trip hazards, moveable items and debris
 - wearing gloves and steel capped boots
 - team lifting
 - using a mechanical aid
 - not twisting under load
 - bending the knees when picking up or placing down materials
 - positioning bricks and blocks to reflect the laying sequence of the project plans
 - positioning bricks and blocks for ease of access and clear of walkways, which may include placing them 500mm from the brick or block wall and 300mm from mortar boards
- Set up, level and stabilise a cement mixer, which may include:
 - finding a flat area, close to mortar materials and water supply
 - placing blocks or chocks against or under the cement mixer to stop it from rolling
 - connecting cement mixer to a residual current device (RCD)
- Set up, level and stabilise a raised mortar board, which may include:
 - finding a flat area, close to the brick or block wall (eg. 500mm) and brick stacks (eg. 300mm)
 - placing bricks, blocks or mortar board folding stand under the mortar board
- Check the brick and block quality and quantity against the delivery docket, which may include:
 - Counting materials and checking they match the delivery docket
 - Checking the quality of materials on the delivery docket
 - Noting and recording details of any material inadequacies or damage on the delivery docket



Brick and Block Coverage

Measure the area requiring bricks or blocks by multiplying the height by the width (from the project plans). In general, the calculation used by bricklayers for standard sized bricks (230mm x 76mm) is 48 bricks per square metre of wall to be built (see example 1). Alternatively, calculate the individual brick surface area (brick length x height, adding 10mm to each measurement for mortar), to determine the required bricks (see Example 2). Always add 10% to your total for cuts and accidental breakages.

Example 1:

The surface area of the following 1.2m x .506m wall is 0.6072 m².

$0.6072 \times 48 = 29$ bricks

29×1.1 (wastage) = 32 bricks

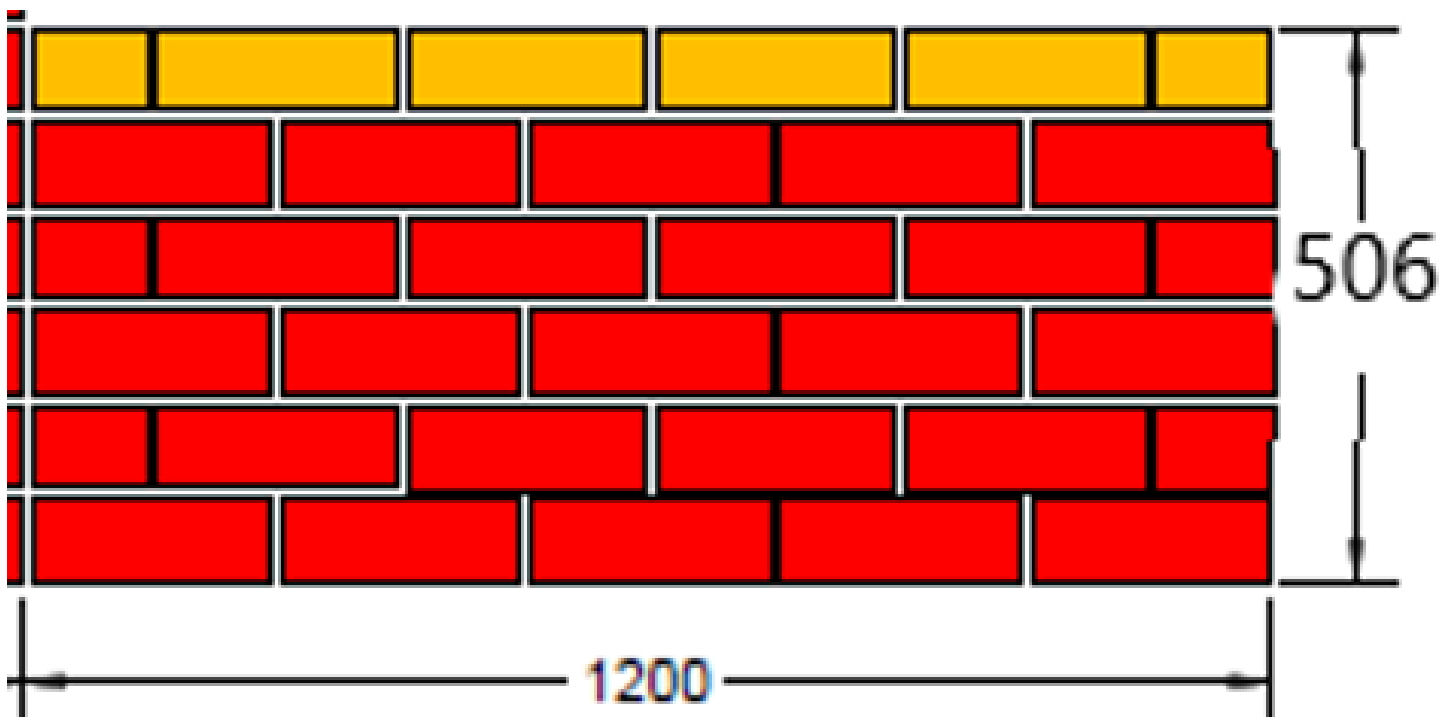
Example 2:

The surface area of the following 1.2m x .506m wall is 0.6072 m².

Each brick (plus 10mm mortar) is 240mm x 86mm (or .24m x .086m), so the surface area of each brick is 0.021m².

Wall surface area $\div$ brick surface area = bricks required ($0.6072 \div .021 = 29$ bricks).

Bricks required plus 10% = bricks ordered ($29 \times 1.1 = 32$ bricks)



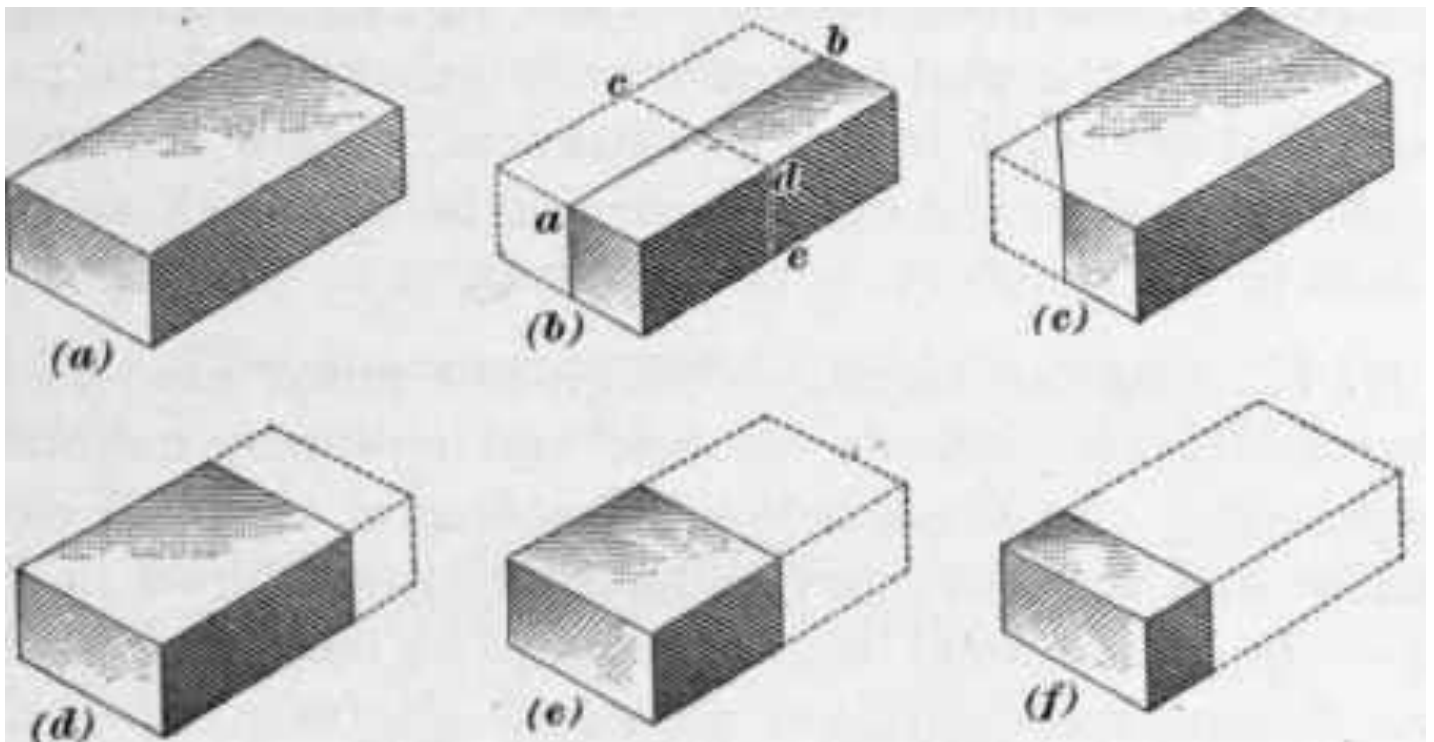
Cutting Bricks and Blocks

Cutting bricks or blocks may be required to achieve a desired shape and size, to work around any obstructions or to form a particular pattern. This can be achieved manually using a bolster chisel and lump/mash hammer. Accurate cutting can be carried out using a wet cut brick saw. Steps to using a wet cut brick saw to cut bricks and blocks may include:

- Reading and interpreting the wet cut brick saw operator manual safe operating procedures (SOP)
- Conducting recommended pre-operational checks, which may include:
 - checking power cords for deformities, cracking and currency of test tags
 - checking hoses and lubricants
 - checking and adjusting guards and guides, controls and switches
 - checking the blade
 - checking the table and roller function
 - checking the foot pedal function
- Setting up, levelling and stabilising the wet cut brick saw, which may include:
 - finding a flat area, close to bricks, blocks and water supply
 - placing blocks or chocks against or under wet cut brick saw to stop it from rolling
 - connecting wet cut brick saw to a residual current device (RCD)
- Follow the SOP to safely cut bricks and blocks to conform with project plans and specifications, which may include:
 - Wearing safety glasses, earmuffs and a mask
 - Turning on the saw and adjusting the water flow
 - Marking out on the brick/block, or for multiple cuts of the same size mark on the table for efficiency
 - Using the pedal to engage the blade so both hands can be used on the table and material being cut
 - Moving the table towards and through the blade to complete the cut
 - Not applying too much force to the material and keep fingers and hands well away from the blade

A New Place Construction employee may be required to use brick and block laying cutting tools to achieve various brick shapes, which may include:

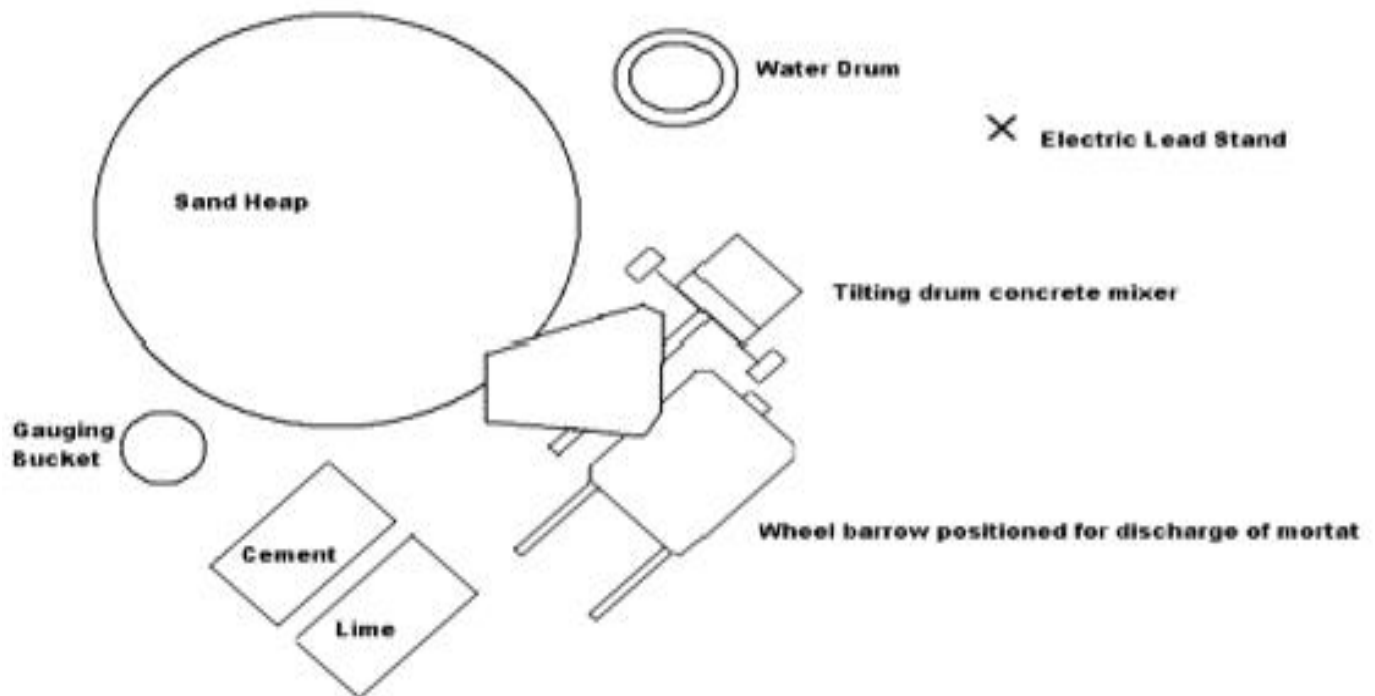
- b) Queen closer
- c) Kink closer
- d) Three-quarter bat
- e) Half bat
- f) Quarter bat



Mixing Mortar

The main function of mortar is to develop a complete and durable bond between masonry bricks, binding them into a single wall. The mortar should be strong enough to withstand load and forces, and it must create a water-resistant barrier. The final mortar mix must be M3 strength, which is a requirement of AS3700 Masonry Structures. The steps in mixing mortar may include:

- Using the bucket gauging technique
- Combining Water : Cement : Lime : Sand in a 2 : 1 : 1 : 6 ratio, always adding water 1st
- Allowing mortar materials time to combine thoroughly
- Confirming the quality and consistency of the mortar with a supervisor



Undertaking Bricklaying

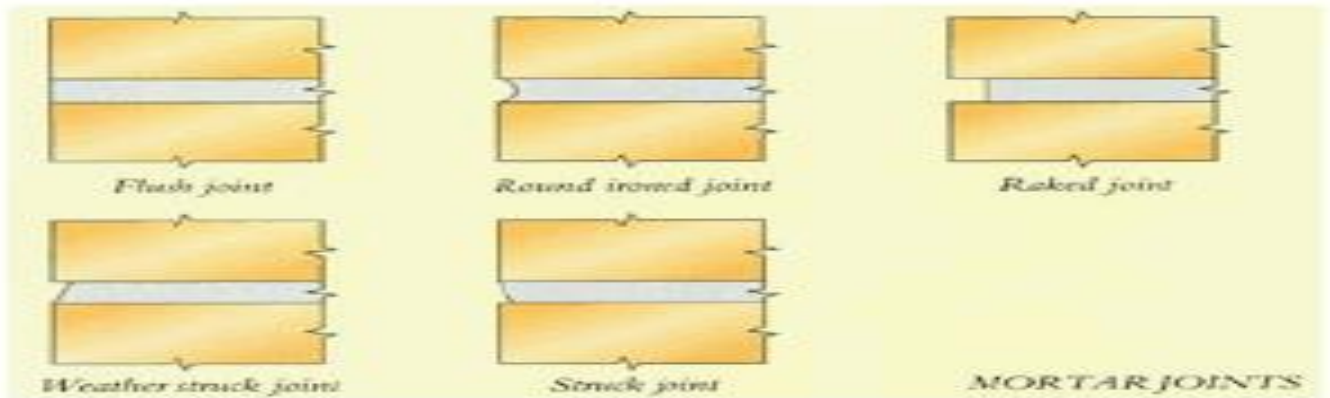
Like all construction activities, it's important to have Standard Operating Procedures. In general, for bricklaying activities, a New Place Construction employee may be required to use bricklaying tools and equipment to achieve a balanced masonry finish that conforms with project plans and specifications, which may include:

- Setting up the plan on the ground to match the aspect of the marked-out bottom course
- Checking foundation or footing levels
- Determine the number of bricks per course
- Setting up a vertical profile at each end of the wall
- Running a string line between the two profiles and using a laser level to set the string line height of the 1st course
- Spreading a 15mm layer of mortar (mortar bed) and furrowing with the trowel
- Placing the corner/end brick in place squarely up against the string line and bed it into the mortar
- Butter up each subsequent brick and bed into the existing laid brick and mortar bed
- Continuing to place bricks gently in line with the previous bricks, up against the string line, using your spirit level to ensure that the wall stays level and true
- Cleaning off excess mortar with the trowel so that it is flush with the bricks
- Once first course is laid, measure lengths as per plan, square pier, use a straight edge against stretcher face and then plumb
- After each course is laid and cleaned, lift the string line up the profiles for the next level to begin, until the wall is complete

Mortar Joint Finishes

Mortar joints finishes are used to accommodate the natural dimensional variations that are seen in clay bricks. Properly filled and tooled or ironed joints improve the durability, weather-proofness and sound performance of brickwork. In general, for jointing activities, a New Place Construction employee may be required to use bricklaying jointing tools to achieve various mortar joints, which may include:

- Flush joint – cut flush with a trowel
- Round ironed joint – using a round iron jointer; used in restoration work on Federation houses.
- Raked joint – using an adjustable raker up to 10mm deep; used in house construction.
- Other common types include struck joints and weather struck joints, made by running the edge of a trowel along the joint.

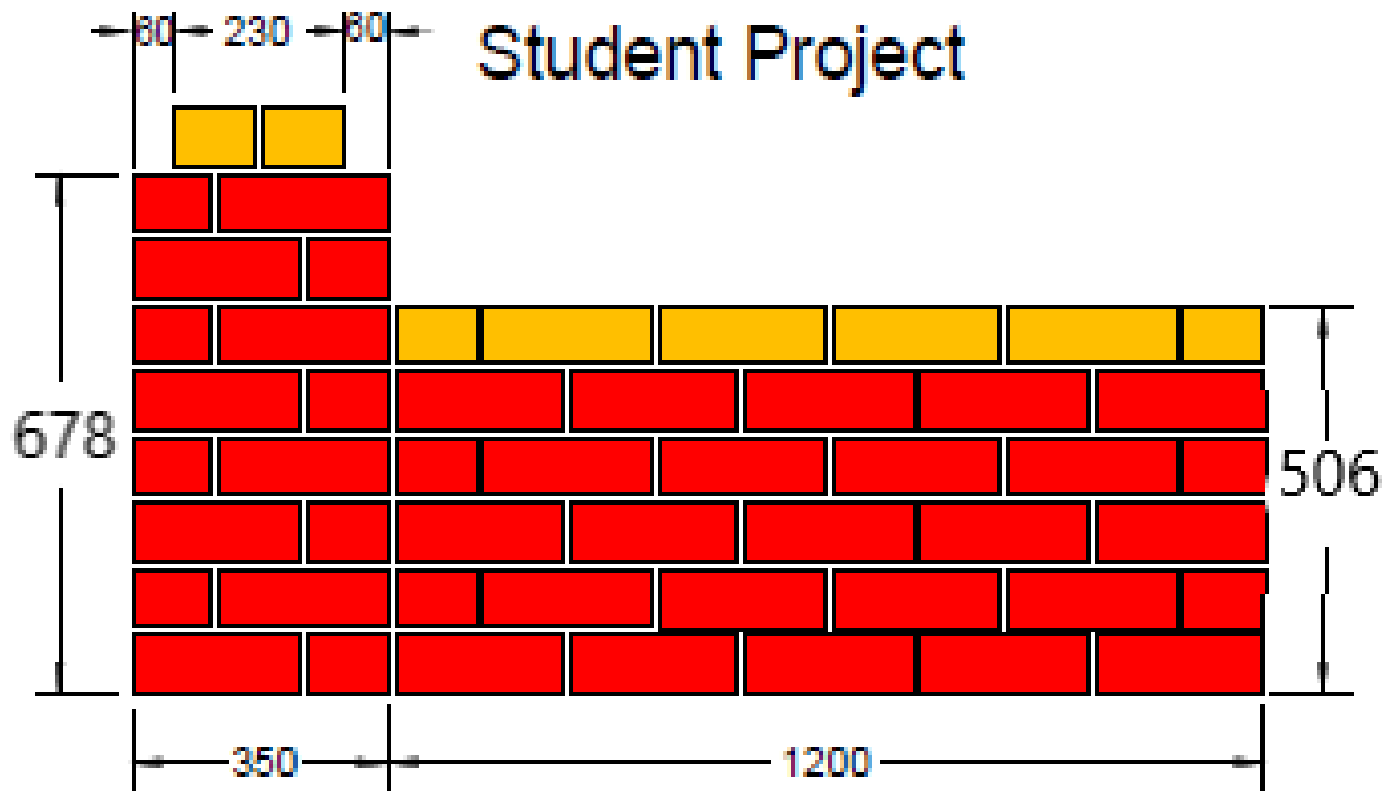


Brick and Block Clean Up

Like all construction activities, it's important to clean-up on the completion of any tasks. In general, for brick and block laying activities, a New Place Construction employee may be required to:

- Clean and clearing the work area, which may include:
 - collecting bricks and blocks that can be reused/recycled and placing them in designated storage area
 - picking up any large clumps of mortar
 - using a shovel to remove brick and block materials and debris
 - placing bricklaying materials and debris into the bin
 - removing bricklaying equipment (eg string line, corner blocks) and place in the designated storage area
- Disconnecting, cleaning and storing portable electric equipment, which may include:
 - unplugging the cement mixer and wet cut brick saw
 - accessing and reviewing the cement mixer and wet cut brick saw manufacturer cleaning and maintenance instructions
 - in a designated wash off area, use water and a brush to clean off mortar and brick residue from the cement mixer and wet cut brick saw
 - storing the cement mixer and wet cut brick saw in a secure and undercover area
- Follow operator manuals to clean, check and maintain tools and equipment, which may include:
 - lubricating moving parts
 - wiping and washing off mortar debris and brick dust in the designated wash off area
 - sharpening bits and blades
 - calibrating equipment
 - replacing worn parts
 - charging battery operated tools
- Store tools and equipment, which may include:
 - checking for faults
 - reporting to the teacher any faulty tools and equipment
 - storing serviceable tools and equipment in the designated storage area

Possible Brick Project 1

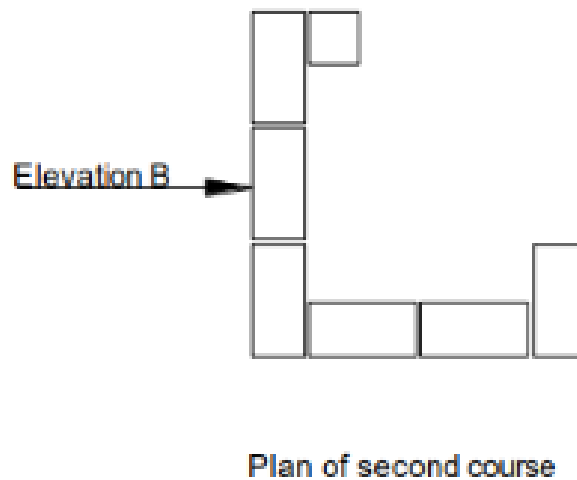
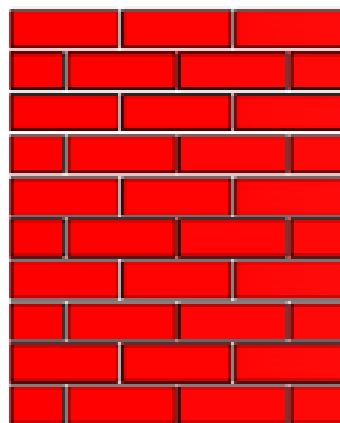
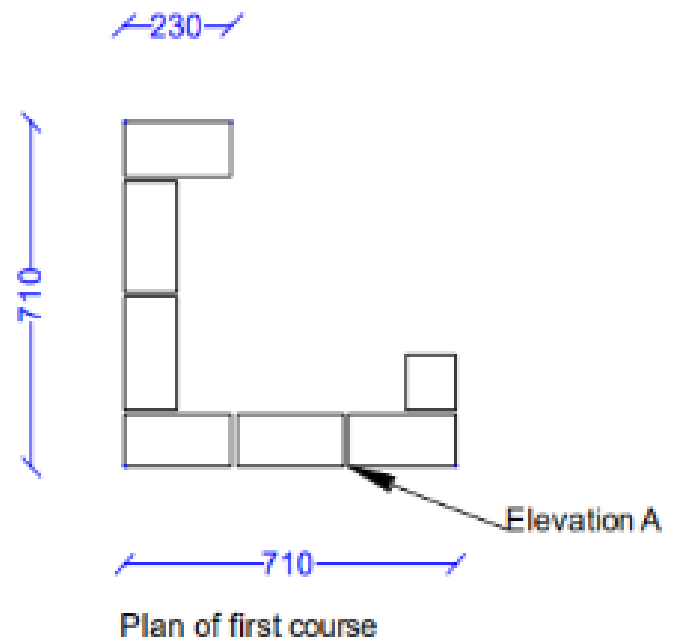
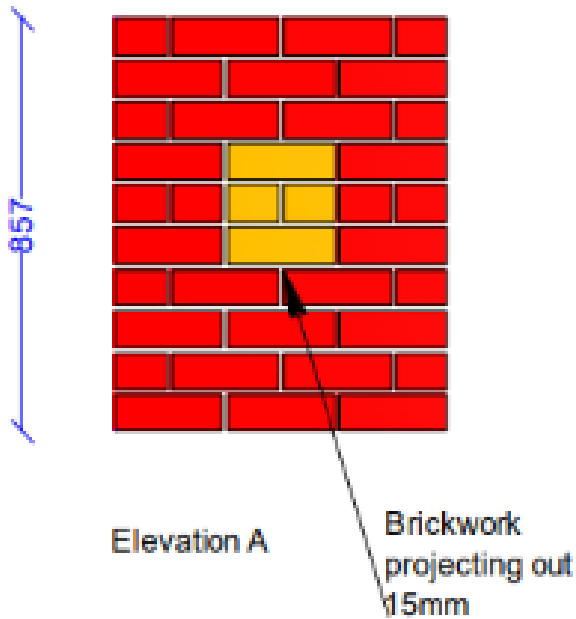


Last course



First Course

Possible Brick Project 2



Possible Block Project

Note: The training package only requires the planning and preparation of a blocklaying project, therefore the blocklaying project does not have to be physically laid by students for assessment.

		390 x 190 x 40		
		390 x 190 x 190		
190 x 190 x 190				

Delivery Docket Template

Date: _____

Docket No: _____

Delivery to: _____

Supplied by: _____

Customer : _____

Tel No: _____

Order Date: _____

Reference: _____

Delivery Date: _____

Date	Description	Quantity		
		Ordered	Shipped	Backorder

Please retain this delivery docket as proof of purchase.

Comments:

